

Explainable Bangla Document Classification Using Attention-Based Deep Learning Models

I. INTRODUCTION

Despite being widely used language on the internet and being spoken by over 230 million people in the world, Bangla (Bengali) is unfortunately one of the most under-resourced languages for natural language processing (NLP) relative to English and other high resource languages. Thousands of news articles are being published daily on Bangla news media on various topics ranging from politics, sports, entertainment, economy and technology. The rapid digitization of Bangla news media has made a need for an automated document classification system to classify, filter and route a huge volume of news articles. Manual categorization is time consuming, varies between editors and does not keep up with the amount of content being created. As such, automated Bangla document classification can be practically used in news aggregation systems, content recommender systems, media monitoring systems, and digital archiving systems.

The grouping of Bangla text, on the other hand, has its own difficulties as compared to the grouping of English documents. Bangla is rich in morphology, having a large number of inflections for tense, cases and honorifics, and has no capitalization marks for the delineation of named-entities or sentences like English. However, frequent code-mixing with English (especially when it comes to informal contexts), inconsistent spelling rules, and a relatively small number of curated, large-scale labeled corpora add further complexity to word segmentation and tokenization. All these facts make it difficult to adapt English-centric NLP pipelines for Bangla and drive the design of preprocessing and modeling options that are explicitly Bangla-specific, which is done in this study by going through the dataset cleaning and tokenization pipeline.

In addition to the pure accuracy of classification, an important demand for NLP systems deployed in practice is *explainability* or explainability. Deep neural networks, especially recurrent and transformer networks, have been often accused of being a “black box” that can make hard-to-explain predictions, make mistakes that are hard to diagnose, or be unreliable in production environments. This is particularly acute for low-resource languages like Bangla - Labeled data is less prevalent, noise is more likely and there are less standard tools available to practitioners for diagnostics. It is hard to say whether the model is truly competent or whether it is using spurious correlations, such as the length of the document, boilerplate wording, or tokens that are relevant to the topic of the document but not part of it and are common in the target language. Explainable AI (XAI) techniques are therefore a new

line of inquiry along with model performance. In addition to model performance, there is therefore a new line of inquiry around “Explainable AI” (XAI) techniques such as attention weights, gradient-based saliency, and post-hoc techniques like LIME and SHAP.

Recent studies have been conducted to implement deep learning and transformer-based models in Bangla text categorization and its associated applications. Transformer models like BanglaBERT, mBERT and XLM-RoBERTa have demonstrated excellent context-modeling performance on Bangla emotion and sentiment corpora [1], [2], while hybrid architectures with both lexicon-based and transformer-based sentiment signals have enhanced robustness against negation and intensity [6]. The Potrika dataset [3] has produced high accuracy results for Bangla news categorization using fasttext based GRU models over traditional classifiers like SVM and Logistic Regression, while the accuracy results are near 100

Together, this literature supports that (i) transformer-based models produce strong baseline performance on Bangla classification and sentiment classification tasks, but are instead evaluated on social media corpora or review corpora or emotion corpora rather than corpora of formal newspaper documents, (ii) variants of recurrent architectures such as GRU models are generally competitive models, and are also often used as a computationally lighter alternative to large transformers, particularly when the amount of labeled data or the amount of computational power is limited, and (iii) explainability techniques like LIME, SHAP, and attention visualization are increasingly paired with Bangla classifiers, but are more often applied to sentiment or emotion tasks than to multi-class newspaper document classification, and are rarely comparatively evaluated across multiple architectures with increasing representational complexity in a single controlled study. Very few studies have (a) benchmarked a lightweight CNN, a BiLSTM and an attention-augmented BiLSTM with the same preprocessing and training setup on a large-scale Bangla newspaper corpus, and (b) used the instance-level, gradient-based explanation mechanism to check whether the predictions are grounded on linguistically meaningful tokens or incidental artifacts.

This paper seeks to fill that void. We present an end-to-end pipeline for multi-class Bangla newspaper document classification that (1) systematically compares the performance of three different neural architectures of increasing complexity – Convolutional Neural Network (CNN), Bidirectional LSTM (BiLSTM) and our proposed Attention-BiLSTM – trained and evaluated with the same tokenization, data splits and

hyperparameter budgets on a corpus of 100,000 articles from the Bangla newspaper domain, split across 9 classes; (2) analyzes the performance of the models using precision/recall/f1-score, confusion matrix and convergence plot for both training and validation not only at the aggregate level but also by class; and (3) audits the decisions made by the best-performing Attention-BiLSTM model with a gradient $\times$ input saliency method, generating instance and token level explanations for individual predictions, providing a practical means to audit model decisions on a low-resource language where such tooling is still limited.

The main contributions of this work are summarized as follows:

- A controlled, reproducible comparison of CNN, BiLSTM, and Attention-BiLSTM architectures for large-scale Bangla newspaper document classification under identical preprocessing and training protocols.
- A detailed empirical analysis identifying which document categories benefit most from an explicit attention mechanism, with particular attention to underrepresented, lexically diffuse categories that are typically hardest to classify.
- An integrated, gradient-based explainability module that produces normalized, token-level importance scores for individual predictions, enabling qualitative verification that the model's decisions align with intuitive linguistic cues.
- An empirical training-dynamics analysis (via early stopping, learning-rate scheduling, and checkpointing) that clarifies the practical trade-offs between model capacity and overfitting on Bangla text of this scale.

The remainder of this paper is organized as follows. Section II reviews related work on Bangla text classification and explainable NLP. Section III describes the dataset, preprocessing pipeline, model architectures, and training protocol. Section IV presents the experimental results and analysis, including per-class performance, confusion matrices, training dynamics, and qualitative explainability findings. Section V concludes the paper and outlines directions for future work.

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